

3.6.2 Food processing and production

3.6.2.1 Food production

Content	Students must know and understand	Suggested application and food preparation skills
primary and secondary stages of processing and production. how processing affects the sensory and nutritional properties of ingredients	• primary processing related to the: rearing, fishing, growing, harvesting and cleaning of the raw food material (milling of wheat to flour, heat treatment of milk, pasteurised, UHT, sterilised and micro-filtered milk) • secondary processing related to: how the raw primary processed ingredients are processed to produce a food product (flour into bread and/or pasta, milk into cheese and yoghurt, fruit into jams) • loss of vitamins through heating and drying • the effect of heating and drying on the sensory characteristics of milk.	• Make dough for pasta, shape and finish dough using a pasta machine, shape and finish pasta (S5 and S10). • Water based cooking methods using the hob to boil the pasta (S6).

3.6.2.2 Technological developments associated with better health and food production

Content	Students must know and understand	Suggested application and food preparation skills
technological developments to support better health and food production including fortification and modified foods with health benefits and the efficacy of these.	• cholesterol lowering spreads • health benefits of fortification • fortified foods: thiamin, niacin, calcium and iron added to white flour • folic acid and iron added to breakfast cereals • vitamins A and D added to fats and low fat spreads • the positive and negative aspects of the use of additives: colourings, emulsifiers and stabilisers, flavourings, and preservatives • the positive and negative aspects of Genetically Modified (GM) foods.	To examine, carry out sensory analysis and evaluate existing products that have been modified and fortified.